This implies

$$\begin{aligned} d(F_i^+V) - d(V) &= -B\left(d(V), \alpha_i\right)\alpha_i \\ \iff \quad d(F_i^+V) \;&=\; d(V) - B\left(d(V), \alpha_i\right)\alpha_i \;=\; s_i\left(d(V)\right). \end{aligned}$$

□

6.7. Coxeter elements

Definition 6.7.1. Let Q be a quiver and let Γ be the underlying graph. Fix any labeling $1, \ldots, n$ of the vertices of Γ. Then the **Coxeter element** c of Q corresponding to this labeling is defined as

$$c = s_1 s_2 \ldots s_n.$$

Lemma 6.7.2. *Let*

$$\beta = \sum_i k_i \alpha_i$$

with $k_i \geq 0$ for all i but not all $k_i = 0$. Then there is $n \in \mathbb{N}$, such that

$$c^N \beta$$

has at least one strictly negative coefficient.

Proof. The Coxeter element c belongs to a finite group W. So there is $M \in \mathbb{N}$, such that

$$c^M = 1.$$

We claim that

$$1 + c + c^2 + \cdots + c^{M-1} = 0$$

as operators on $\mathbb{R}^n$. This implies what we need, since β has at least one strictly positive coefficient, so one of the elements

$$c\beta, c^2\beta, \ldots, c^{M-1}\beta$$

must have at least one strictly negative coefficient. Furthermore, it is enough to show that 1 is not an eigenvalue for c, since

$$\begin{aligned} (1 + c + c^2 + \ldots + c^{M-1})v = w \neq 0 \\ \implies \quad cw &= c\left(1 + c + c^2 + \cdots + c^{M-1}\right)v \\ &= (c + c^2 + c^3 + \cdots + c^{M-1} + 1)v = w. \end{aligned}$$